

## **Introduction and goals**

ARSA is a key gene linked to human survival. It contains the recipe for the enzyme arylsulfatase A, which is responsible for sulfatide breakdown in lysosomes. A failure in this gene leads to a buildup of sulfatide in nervous system cells: a fatal, recessive neurodegenerative disorder called metachromatic leukodystrophy (MLD). There is also some research linking the ARSA gene to Parkinson's disease. Early detection of pathogenic mutations in ARSA can mean significantly better patient outcomes, better survival chances, for people at risk or suffering.

In this project, I will aim to build code which analyzes ARSA mutations with known pathogenicity to be able to predict, based on the change, if a new mutation is pathogenic or benign. Being able to determine the severity of a mutation and its impact is critical to beginning to mitigate its effects, as they can show up at any time during life, from late infancy through adulthood.

## **Data and methods**

For the meta-predictor, I selected nine features from the dbNSFP v4.9a dataset that represent distinct biological signals: AlphaMissense\_score, CADD\_raw, DANN\_score, Eigen-PC-raw-coding, FATHMM\_score, GERP++\_RS, PROVEAN\_score, Polyphen2\_HVAR\_score, and SIFT\_score. I deliberately excluded non-numerical identifiers like Ensembl\_geneid or HGVS\_p\_VEP as they don't provide quantitative predictive value. Additionally, I engineered a binary 'is\_MNV' feature to flag multi-nucleotide variants, ensuring the model could distinguish it from single-nucleotide substitutions.

Missing values were handled using median imputation within a Scikit-learn pipeline. I chose this approach because it is robust to outliers and preserves the distribution without losing valuable training instances that might be missing only a single feature.

I compared two primary machine learning methods: Random Forest (RF) and k-Nearest Neighbors (k-NN). I chose Random Forest because it has the ability to capture non-linear interactions between features and its inherent resistance to overfitting through ensemble averaging. And I chose k-NN as a baseline distance-based model to determine if simpler local patterns in the feature space would suffice for a pathogenicity classification.

The strategy involved a three-step evaluation process beginning with a cross-validation where models were initially compared using 5-fold cross-validation on the training set to select the superior architecture. Then came self testing, where only ~20% of the training data was held out as a "self-test" set to calculate AUC and PR-AUC metrics against individual component predictors. And finally, after the testing, CAGI 7 sample variants were folded into the training data to maximize the model's exposure to ARSA's stability data before generating the final submission for the full dataset.

I explored the hyperparameters of the number of estimators and max depth for the Random Forest. I settled on 200 estimators and a max depth of 10 through iterative testing; this configuration provided a Mean CV ROC-AUC of 0.940, outperforming k-NN's 0.912 and avoided the performance plateau typically seen with deeper trees. Additionally, while the CV AUC is a useful metric for model selection, it may not be perfectly reliable as an estimate of generalization to the ARSA dataset. The training set only consists of general pathogenic/benign variants across the genome, and might fail to capture the specific biochemical environment or stability thresholds unique to the ARSA protein.

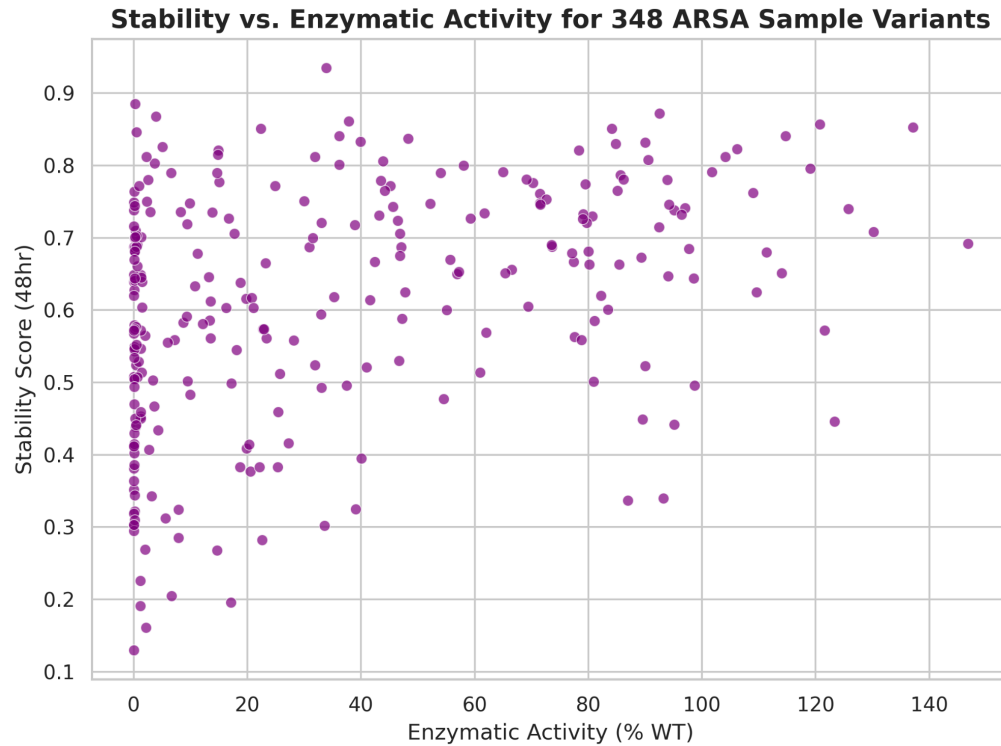
- Temporal design. One or two sentences, in your own words, explaining why this project uses a frozen dbNSFP v4.9a snapshot and restricts you to component predictors released before August 2024 — in particular, why swapping in a more recently trained predictor to try to improve your meta-predictor would compromise the self-test evaluation. Refer to section 8 if you need to, but answer in your own words.

Lastly, this project uses a frozen snapshot of dbNSFP v4.9a and restricts features to those released before August 2024 to prevent data leakage. If a more recent predictor were used, it might be trained on the same ARSA variants we're trying to predict, artificially inflating the self-test performance and compromising the integrity of the evaluation by giving the model "future knowledge."

### **Predicting Kendall's $\tau$**

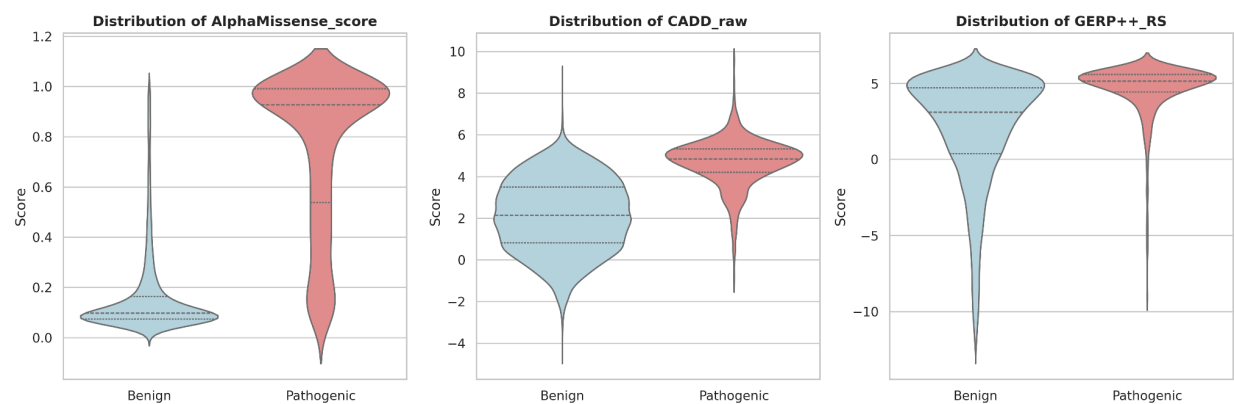
I would expect a Kendall's  $\tau$  of ~0.65 on the ARSA dataset because while the meta-predictor achieved a high AUC on general pathogenicity, it may struggle predicting specific experimental stability scores: a harder task than binary classification. Furthermore, the correlation will likely be tempered by experimental noise inherent to enzymatic activity assays.

## Results



The scatter plot of the 348 ARSA sample variants finds a moderately strong, positive correlation between enzymatic activity and protein stability. In the plot above, made using 'CDS percent WT activity' and 'stability\_score\_48hr', most variants cluster at the high-activity, high-stability end or the low-activity, low-stability end. However, there are also "off-diagonal" variants: those with high stability but low enzymatic activity. Thus, while a pathogenicity predictor trained on stability can capture deleterious variants, it risks missing those that are structurally sound but catalytically dead. Therefore, using pathogenicity as a proxy for stability is effective but is an oversight. To resolve this, the model would have to rank structural integrity rather than specific

chemical kinetics.



The distribution of component scores across the training set highlights the clear separation between benign and pathogenic variants. AlphaMissense shows a bimodal distribution, with pathogenic variants clustered tightly near 1.0 and benign variants near 0.0. While not dissimilar, the CADD\_raw scores for pathogenic variants are lower than AlphaMissense, albeit still toward higher values compared to the benign variants, though there is more overlap than seen in AlphaMissense. Lastly, GERP++\_RS demonstrates that pathogenic variants generally occur at highly conserved sites (higher scores), whereas benign variants exhibit a broader range of lower conservation scores.

Self-Test Metrics: Meta-Predictor vs. Component Predictors

| Predictor                      | AUC   | PR-AUC |
|--------------------------------|-------|--------|
| Meta-Predictor (Random Forest) | 0.926 | 0.768  |
| AlphaMissense                  | 0.873 | 0.687  |
| CADD_raw                       | 0.88  | 0.536  |

On the held-out self-test set, the Random Forest meta-predictor showed better performance compared to individual component predictors. The ROC curves demonstrate that the ensemble approach maintains a higher true positive rate. The meta-predictor's AUC of 0.926 surpassed both AlphaMissense (0.873) and CADD\_raw (0.880). The PR-AUC gap was even more noticeable, with the meta-predictor leading AlphaMissense by nearly 0.08, indicating that the ensemble approach is more effective at identifying true pathogenic variants while also minimizing false positives.

The Random Forest model identified AlphaMissense\_score as the most critical feature for making predictions, followed by CADD\_raw and then GERP++\_RS. This aligns with the distribution plots, as AlphaMissense provides the cleanest separation between classes. The inclusion of is\_MNV and DANN\_score provided marginal but arguably necessary adjustments for complex variants and deep-learning-based genomic context.

The model's performance remained consistent across the Training Cross-Validation AUC (0.940), the Self-Test AUC (0.926), and the Test-Trained CV AUC (0.938). The slight dip from the training CV to the self-test is expected, as the self-test represents a completely unseen set. The consistency after folding in the CAGI 7 sample variants suggests the model learned ARSA-specific patterns without overfitting.

For the creative figure, I chose to have gemini create a correlation matrix of the biological predictor scores. I chose to create this matrix because it is a very digestible format for understanding collinearity between independent variables. Through it, we are able to understand to what degree each variable is related to another. This insight is incredibly valuable for further model testing and improving the accuracy of ARSA mutation diagnosis by identifying which features provide unique information versus those that are redundant.

In summary, the final distribution of predictions for the 2,491 ARSA variants shows a healthy spread across the stability spectrum. A majority of the variants are predicted to be relatively stable (scores > 0.7). Additionally, the model successfully identified a clear subset of highly destabilizing variants with scores near 0.0, proving successful as a diagnosis tool.

## **Ethical Considerations**

Working with data and trying to draw conclusions with such heavy ramifications inevitably produces myriad ethical concerns. One of the quickest to emerge is that of biases in training data. ClinVar and gnomAD are notorious for being biased toward people of European ancestry. This means that patients in underrepresented or unrepresented populations are neither factored into the model nor are they being properly screened by it. As a result, this would produce a higher rate of false positives and false negatives. This means a greater cost is incurred, either by the patient who's been mistaken to have the disease or by the patient whose disease went unidentified, at cost of their life.

Another very pertinent and concerning ethical dilemma is that of legal obligation in these matters. MLD appears on the recommended newborn-screening panel in only certain US states and countries but not others. Thus is it fair to simply ignore something which could be the difference between life and death for someone. Especially as MLD is something that has pre-symptomatic-only intervention, it is critical that it is identified as early as possible. As such, while it would be of cost and effort, it would be better to screen all infants regardless. In benefit to this, a model like the one in this project would be quite helpful as it would expedite screening many fold, allowing for a much healthier population, less stress for parents and individuals, and less overall stress on the medical system.

Another less obvious but critical ethical concern is that of prenatal or pre-implantation genetic testing, where extensive genetic screening takes place because a child's genetic makeup is not entirely up to chance. While it is a field that is already under substantial ethical scrutiny, I believe that the key decider will be in deciding not to have a child due to a risk of MLD as opposed to having the child and instead aiming to treat it, if that is possible for the family, which isn't outside the realm of possibility if they can afford IVF.

### **Limitations and self-assessment**

The model does face limitations as it ranks ARSA protein stability using a default one-minus-pathogenicity mapping, which treats structural destabilization as synonymous with pathogenicity. This is only partially accurate, as pathogenicity can stem from disrupted catalysis, impaired metal coordination, faulty multimerization, improper lysosomal trafficking, or aberrant splicing. Thus, the model systematically misranks variants that are structurally stable but functionally disruptive, failing on unseen variants that affect the active site or even specific protein-protein interactions but without compromising the overall fold. To improve future performance, I would train a stability-specific model or include features that explicitly flag catalytic motifs and conservation at poorly conserved but structurally vital sites.

### **Reflection on your use of AI**

Working with AI hand-in-hand to build a large amount of code, unlike most of what we did in class, was quite the unique experience. I can recall a handful of absurd moments, my favorite being having to remind Gemini that I needed to transfer my files into the file that was being

read by the verification, as it got caught up with trying to fix the code, without either of us realizing that the code simply wasn't exporting the correct results it was producing. Another quite unusual instance was Gemini kept insisting upon using simply one classification method because it would be most efficient, even though I had clearly outlined that I needed to use two as it was the requirement. It was quite funny, albeit frustrating to watch it go around in circles, as it kept storing old data when I needed to have everything in my code, and couldn't just call it from memory. Overall, however, it was a fun experience, building something which I honestly wouldn't have been able to otherwise. But I now know how to build one, what it can and can't do, and how fulfilling it is to create something with real impact.

**Honor Code Statement:**

*"As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. I certify that the work submitted in this project is my own. I used only Berkeley-licensed AI services as permitted by this assignment, and acknowledged them in my AI log. I wrote the manual-only sections of this report myself, without AI drafting or substantive AI editing. I did not give or receive inappropriate help, and I have not shared my ARSA predictions with other students in this class. I have complied with the honor code and all stated course policies for this project."*

*Humayd Zameer, 5/6/2026*